

KERNEL

Core program that talks directly w the hardware

SHELL

Reads commands and execute them

PROCESS

Running instance of a program

DAEMON

Background service processes with no terminal attached

SSH

Secure remote shell access

FILE STRUCTURE & BASIC COMMANDS:

/: root

/home/xyz: my home

/etc: system wide config files, settings that control how programs behave

/var: variable data, logs mails, db, cache

/usr: installed programs+files

/bin: executable files(every basic command like ls, cat etc, exists as a physical file)

```
zarmeena@z:~$ which ls
/usr/bin/ls
zarmeena@z:~$ ls -l $(which ls)
-rwxr-xr-x 1 root root 142312 Aug 25 15:09 /usr/bin/ls
```

/tmp: temporary files, wiped on reboot

/root: root users home directory

/dev: device files, hardware stuff

/proc: live kernels, process informations

Touch: make new files

Mkdir: make new directories

Mkdir -p parent/child/grandchild (nested directories all at once)

Cp: to copy

Cp source.txt destination.txt

Cp -r directory1 directory2 (recursive copy)

Mv: to rename or to move

Mv oldname.txt newname.txt

Mv filename.txt directory

Rm: to delete

Rm file.txt

Rm -r directory (delete directory and everything in it)

Rm -rf directory (forceful deletion)

Find: searches for files by name

Find directory -type f/d -name/type/mtime "pattern"/time by day

Pattern: *pattern. Anything ending w pattern,

Is -la: all files in the directory, including hidden files

Is | head -20: list first 20 lines

'|' takes output of one command and feeds it into another. In this case output from 'ls' is placed into command head -20

Ls | tail -20: last 20 lines

Ls | head -30 | tail -10: take last 10 of the first 30 lines

Ls | sed -n '20,30p'

-n suppresses default option, by default sed prints every line it processes

'20,30p' to print(p) line 20 to 30

Ls | sed -n '\$p': prints last line(\$)

sed : stream editor

Uses: find and replace text, print specific range, delete lines matching a pattern

Sed 's/old/new' , by default only replaces the first match

Sed 's/old/new/g' g for global, makes change globally

Sed -i 's/old/new/' filename edits change into the file

Sed -i '5d' filename , deletes line 5 from filename

Sed -i '/pattern/d' filename , deletes line containing pattern

Sed -i '5,10d' filename , deletes line 5 to 10 in file

Sed -i 's/pattern/PATTERN/gi' filename , case insensitive deletion

s: substitute

old: old pattern to find

new: replacement

~/home/xyz

~ points to home directory

Ps: process status, currently running processes on da system

Set of flags

aux: 'a'=all users, 'u'=user oriented format(owner, %cpu), 'x' = processes w/o controlling terminal(background/daemon processes)

-ef: same as aux, shows full ppid (parent process id)

-u zarmeena : processes owned by me

-e: minimal columns(PID, TTY(terminal),TIME CMD)

-eo pid,%cpu: custom columns

-aux -sort='%cpu' : sort by %cpu in descending order

-T: only processes tied to current terminal

Ps aux | sort -k3 -nr | head -5

-k3: to sort based on column3

-n: if not used, sort will treat values as text, arranging alphabetically, tells sort to compare them numerically

r: sort by default is in ascending order, this reverses it

Grep: filters lines matching pattern

Grep "pattern" filename (prints everyline in filename containing pattern)

Command | grep "pattern"

Flags (grep <flag> pattern)

- i: case insensitive
- v: invert, show lines that dont match
- n: show line numbers alongside matches

```
zarmeena@z:~$ ps aux | grep ssh
zarmeena  1472  0.0  0.1  6544  2356 tty1    S+   10:43   0:00 grep --color=auto ssh
zarmeena@z:~$ ps aux | grep -n ssh
116:zarmeena  1474  0.0  0.1  6544  2356 tty1    S+   10:44   0:00 grep --color=auto -n ssh
```

- r : recursive, search all files in da folder or subdirectory
- c: count matches instead of printing them
- w: match the whole word only 'cat' instead of 'category'
- A 3: show 3 lines after each match
- B 3: show 3 lines before each match

Grep + regex powers

Grep "^pattern" filename: displays lines where pattern is at start of line

Grep "pattern\$" filename: displays lines where pattern is at end of line

Grep -E "pattern1|pattern2" filename: displays lines where either pattern 1 or 2 is available

Awk: column based text processing

Awk splits each lines into fields by whitespaces by default and refers to them as \$1, \$2 etc where \$0 is the whole line

Awk '{print \$column}' filename

Command | awk '{print \$column}'

```
zarmeena@z:~$ ps aux | awk '{print $1, $2, $4}' | head -5
USER PID %MEM
root 1 0.6
root 2 0.0
root 3 0.0
root 4 0.0
```

Awk -F: '{print \$column}' filename

-F is custom field separator which in this case uses ':' to separate

Use awk for conditional filtering

Ps aux | awk '\$3>1.0 {print \$1, \$3}': only print processes where %cpu is greater than 1

Built in variables

NR(current line no.), NF (no. of fields in current line)

Ps aux | awk '{print NR, \$0}'

awk -F: '\$7 ~ /bash/ {print \$1}' /etc/passwd

\$7 ~ /bash/ matches regex pattern, checks if field 7 contains bash anywhere

cd .. : go one step up in the tree

Cd - : go back (like the back button on browser)

cd/cd ~: go home

Cat: concatenate, basic way to output file content

Cat filename.txt

Cat filename1.txt filename2.txt (prints content of both)

Cat filename1.txt filename2.txt > mergedfile.txt (copies content of both files to new file)

Cat >> file.txt (goes into append mode, type into file directly from terminal , > over writes the file, >> appends it)

Stat filename: for metadata

Wc: wordcount

Wc -l: counts lines

Wc -w: counts words

Wc -c: counts bytes

Wc -m: counts characters

PERMISSIONS

```
drwxr-x--- 4 zarmeena zarmeena 4096 Sep  8 09:30 .
```

drwxr - x - - -

1. First place 'd'
First place is for file type: 'd' for directory, '-' for file
2. Rwx: Second,third,fourth place
For owner permission: read,write,execute. Currently owner has it all
3. R-x: fifth,sixth,seventh place
For group permission: read and execute only
4. - - - : eighth, ninth, tenth place
For others: nothing is allowed

Dif btw write and execute:

Write modifies content (edit,overwrite contents of a file, create,delete,rename files inside directory)

Execute runs program/script

(Without x on a folder, you can't even ls a file inside it by direct path, even if you have r on the folder itself, because r on a directory only lets you list names, x lets you actually traverse into it and access things)

Chmod: change mode - for permissions

Chmod 000: no one has any permission

Chmod 777: everyone have all the permissions

1 for read, 2 for write, 4 for execute

Via Symbols

u/g/o/a (user,group,others,all)

With +/-/= (add,remove,set)

Chmod u+x

Chmod u=rwx,g+w,o-wx

Difference btw + and =

'+' adds permission

'=' sets permission, removing anything that is not mentioned. Chmod o=r, will remove writing and executing permissions, if they had it beforehand

/etc/passwd

username:x:UID:GID:comment:home-dir:shell

1 2 3 4 5 6 7

2- x means that the real password is stored elsewhere (in shadow for security)

/etc/shadow

Hashed password, only accessible to root user

/etc/group

groupname:x:GID:member1,member2

Sudo useradd -m xyz

-m creates their home directory

Sudo passwd xyz

get prompted to give user password

```
zarmeena@z:~$ id tester
uid=1001(tester) gid=1001(tester) groups=1001(tester)
zarmeena@z:~$ cat /etc/passwd | grep tester
tester:x:1001:1001::/home/tester:/bin/sh
```

Id xyz gives UID,GID

Cat /etc/passwd | grep xyz verifies users existence

Sudo userdel -r xyz

Deletes user

-r deletes user's directory

sudo usermod -aG sudo xyz

-aG: a is for append, G specifies the group, -G on its own overwrites

Appends xyz to sudo group

Su xyz

Switches to xyz

Sudo groupadd <group name>

Cat /etc/passwd | grep <group name> (wont show any users)

Sudo usermod -aG <group name> xyz

Groups xyz

Sudo gpasswd -d xyz <group name> (remove user from group)

Sudo groupdel <groupname> (delete group)

Killing process user is in

Ps -u xyz (check first)

Sudo pkill -u xyz(kill-polite command)

Sudo pkill -9 -u xyz (kill-mandated)

DISK MANAGEMENT:

lsblk: list block devices, shows every disk and partition in tree view

```
zarmeena@z:~$ lsblk
NAME
sda
├─sda1
├─sda2
├─sda3
└─ubuntu--vg-ubuntu--
sr0
```

Sudo fdisk -l: lists detailed info like size, types, sectors, boundaries etc

Disk usage

df -h: 'disk free' human readable(gb instead of bytes)

Du -sh /var/log: 'disk usage' -s summary (just total) -h human readable

```
zarmeena@z:~$ sudo du -sh /var/log
27M    /var/log
```

Du -sh /var/log/*

Shows disk usage per item inside

Mounting

Cat /etc/fstab

Shows what gets mounted at every boot

Mount

Shows everything currently mounted right now

Sudo mount source-device target-device

Swap Space

Free -h

Shows ram and swap usage together

What is swap?

Disk storage that acts as overflow for RAM, system low on RAM, kernel moves less active page to swap, then moves back when needed. Acts like safety net. It is an emergency buffer.

Reading and writing to swap is extremely slow

Sudo swapon --show

Shows active swap devices

```
zarmeena@z:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           1.9Gi          341Mi        1.4Gi         1.1Mi        350Mi        1.6Gi
Swap:          2.0Gi          0B          2.0Gi
zarmeena@z:~$ sudo swapon --show
NAME      TYPE      SIZE USED PRIO
/swap.img file      2G    0B   -2
zarmeena@z:~$ ls -lh /swap.img
-rw-r--r-- 1 root root 2.0G Sep  8 08:45 /swap.img
```

SERVERS

Ssh

```
PS C:\Users\JOHN> ssh zarmeena@10.0.2.15
ssh: connect to host 10.0.2.15 port 22: Connection timed out
```

Currently the vm is isolated from host machine, set up port forwarding

```
PS C:\Users\JOHN> ssh zarmeena@127.0.0.1 -p 2222
kex_exchange_identification: read: Connection reset
Connection reset by 127.0.0.1 port 2222
```

Connection reached the port but got immediately reset

Started ssh server on linux

Firewall basics

Ufw: uncomplicated firewall

Sudo ufw status

Sudo ufw allow 22/tcp

Sudo ufw allow OpenSSH

Sudo ufw enable

sudo ufw status verbose

HOSTING SOFTWARE

Every background service on linux server is managed by system-md

Sudo systemctl status

cat /lib/systemd/system/ssh.service

System file that tells systemd how to run ssh

Systemctl list-units --type=service --state=running

Shows every active service on system

Journalctl -u ssh

Logs for ssh

Journalctl -u ssh -f

-f=follow

Live tails the log in real time

The kernel's job ends once its loaded into memory, from that point onwards, one single process (PID1) takes over, responsible for mounting, starting enabled services, ending orphan processes. Systemd keeps monitoring the service, looking out for crashes.

Live monitoring
top/htop

parent/child relationship

Every process except PID 1 has a parent.

Zombie process

A process finishes running but doesn't disappear. It sends exit status to the parent and is now 'technically' dead(no using resources, running) and is waiting for the parent to acknowledge the exit.

Zombie process can't be killed. Only dies when parent is restarted or dies.

Orphan process

A process that is still running even though its parent died(crashed, ended). In this case, the kernel reparents the orphan to pid 1

Mimicking orphan process

```
zarmeena@z:~$ sleep 300 &
[1] 25290
zarmeena@z:~$ bash -c 'sleep 60 &'
zarmeena@z:~$ ps -ef | grep sleep
zarmeena  25290    1012  0 16:54 tty1    00:00:00 sleep 300
zarmeena  25292      1  0 16:54 tty1    00:00:00 sleep 60
zarmeena  25294    1012  0 16:54 tty1    00:00:00 grep --color=auto sleep
```

Cronn: for job scheduling

Crontab -e { to edit }

Crontab -l (to list current cron jobs)

* * * * * jobtorun

minute(0-59)-hour(0-23)-dayofthemoth(1-31)-month(1-12)-dayoftheweek(0-7,0 and 7 being sunday), * means every

*/15 * * * *: every 15 mins

linuxjourney.com

overthewire.org/wargames